

Monday Reading Assessment: Unit 5, Power Consumption

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1 Memory Bank

1. $\Delta E = P\Delta t$... Definition of power in terms of energy consumption, ΔE , and time duration, Δt
2. 1 calorie = 4.184 Joules
3. 1 kcal = 4184 Joules
4. $\Delta U = mg\Delta y$... Change in gravitational potential energy

2 Power Consumption

1. (a) What is the average useful power output of a person who does 6.00×10^6 J of useful work in 8.00 hours?
(b) Working at this rate, how long will it take this person to lift 2000 kg of bricks 1.50 m to a platform? (Work done to lift his body can be omitted because it is not considered useful output here.)

Table 7.4 Basal Metabolic Rates (BMR)

Organ	Power consumed at rest (W)	Oxygen consumption (mL/min)	Percent of BMR
Liver & spleen	23	67	27
Brain	16	47	19
Skeletal muscle	15	45	18
Kidney	9	26	10
Heart	6	17	7
Other	16	48	19
Totals	85 W	250 mL/min	100%

Table 7.5 Energy and Oxygen Consumption Rates^[2] (Power)

Activity	Energy consumption in watts	Oxygen consumption in liters O ₂ /min
Sleeping	83	0.24
Sitting at rest	120	0.34
Standing relaxed	125	0.36
Sitting in class	210	0.60
Walking (5 km/h)	280	0.80
Cycling (13–18 km/h)	400	1.14
Shivering	425	1.21
Playing tennis	440	1.26
Swimming breaststroke	475	1.36
Ice skating (14.5 km/h)	545	1.56
Climbing stairs (116/min)	685	1.96
Cycling (21 km/h)	700	2.00
Running cross-country	740	2.12
Playing basketball	800	2.28
Cycling, professional racer	1855	5.30
Sprinting	2415	6.90

Figure 1: (Top) Metabolic rates of various physiological systems. (Bottom) Power consumption of various human activities.

by these two activities?

2. Consider Fig. 1 (top). (a) What fraction of total power consumption is required by the human brain?
(b) How many Joules of energy does the brain require in a day (estimate)? (c) Convert this to calories, then kcal.

4. A large household air conditioner may consume 15.0 kW of power. What is the cost of operating this air conditioner 3.00 hours per day for 30.0 days if the cost of electricity is \$0.110 per kilowatt hour?

3. Consider Fig. 1 (bottom). (a) If you are caught in the cold, and begin to shiver, how many Joules of energy would you lose in 30 minutes? (b) Compare this to the amount of energy required to sit at rest (awake) for 30 minutes? (c) What is the ratio of energies consumed